

① An image appears too bright.

When an image appears too bright, the intensity values are concentrated in the higher gray-level range, making details difficult to observe. The most suitable gray-level transformation is the Power-Law (Gamma) Transformation with $\gamma > 1$, because it reduces the brightness while preserving image information.

The Power-law transformation modifies the intensity using the equation:

$$S = C \times r^\gamma$$

where,

r = input pixel intensity

S = output pixel intensity

γ = gamma value

C = constant

where $\gamma > 1$, high-intensity pixels become darker.

Application process;

- ① Read the Image
- ② Select gamma value greater than 1.
- ③ Apply the transformation to every pixel.
- ④ Generate the enhanced image.

Expected Outcome

- Overall brightness decreases.
- Hidden details become visible.
- Image appears more natural.
- High-intensity regions are preserved without saturation.

Justification

This method effectively reduces excessive brightness while maintaining image quality and preserving important features.

3) A dark image lacks details

Suitable Grey Level Transformation: Power-law
(gamma) Transformation ($\gamma < 1$)

$$S = c \times r^\gamma$$

when $\gamma < 1$, low-intensity pixels are stretched toward higher intensity values.

Application process:

- 1) Read the image
- 2) choose gamma less than 1.
- 3) Apply transformation every pixel

Expected Outcome:

- * Dark Correction regions become brighter.
- * Hidden objects become visible
- * Bright regions remain almost unchanged.

→ Gamma correction is preferred because it selectively enhances darker regions without over-brightening the entire image.

③ An image has poor contrast :

→ when an image has poor contrast pixel intensities occupy only a small range.
Histogram Equalization spreads the intensity values across the full gray-level range.

Working process :

- ① Calculate histogram
- ② Compute cumulative histogram (CDF)
- ③ Normalize the CDF.
- ④ map old intensity values to new values.
- ⑤ Generate enhanced image.

Expected Outcome :

- Contrast ~~Increases~~
- Image details become clearer.
- Edges become more visible.

A low-contrast image requires uniform intensity distribution.

To achieve a uniform intensity distribution, histogram equalization is the most appropriate method because it redistributes pixel intensities across the entire gray-level range.

working process.

- ① Compute Image histogram
- ② Calculate Cumulative probability.
- ③ Normalize cumulative values.

Effect on Intensity Levels:

- Pixel values spread uniformly.
- Dark regions become brighter
- Bright regions gain more detail
- Overall Contrast Improves

Significantly.

① An Image shows uneven brightness across regions:

when different regions of an Image have different levels, contrast limited adaptive histogram Equalization is the best enhancement technique.

working process:

- ① Divide the Image into small regions.
- ② Perform histogram equalization on each tile.
- ③ Limit Contrast using a clip limit.

Expected Outcome:

- * Corrects non-uniform Illumination
- * Improves Local Contrast
- * Preserves Image details
- * Reduces noise amplification
- * Reduces a natural-looking enhanced Image.